

Inverse variation



- 1 If y varies inversely with x , write an equation that uses k as the constant of variation.
- 2 If $(10, 15)$ is on the curve $P = \frac{k}{Q}$, solve for k .
- 3 Determine whether each of the following is an example of a direct variation or an inverse variation:
 - a The variation relating the distance between two locations on a map and the actual distance between the two locations.
 - b The variation relating the number of workers hired to build a house and the time required to build the house.
- 4 If r varies inversely with a , write an equation that uses k as the constant of variation.
- 5 Determine whether the following equations represent an inverse relationship between x and y :
 - a $x = 1 + y^3$
 - b $x = \frac{8}{y^2}$
 - c $y = 6x + 8$
 - d $xy = -7$
 - e $x = \frac{2}{y}$
 - f $xy = 5x$
- 6 Consider the equation $s = \frac{375}{t}$.
 - a State the constant of proportionality.
 - b Find the exact value of s when $t = 6$.
 - c Find the exact value of s when $t = 12$.
- 7 State whether the following tables represent an inversely proportional relationship between x and y .

a

x	1	2	3	4
y	3	1.5	1	0.75

b

x	1	2	3	4
y	36	18	12	9

c

x	1	5	6	10
y	3	75	108	300

d

x	1	2	3	4
y	4	5	6	7

- 8 m is proportional to $\frac{1}{p}$. Consider the values in the table which represents the relationship.

p	4	6	7	x
m	63	y	36	28

- a Determine the constant of proportionality, k .
- b Find the values of x and y .
- 9 Find the equation relating t and s for each of the following tables of values:

a

s	1	2	3	4
t	48	24	16	12

b

s	3	6	9	12
t	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$

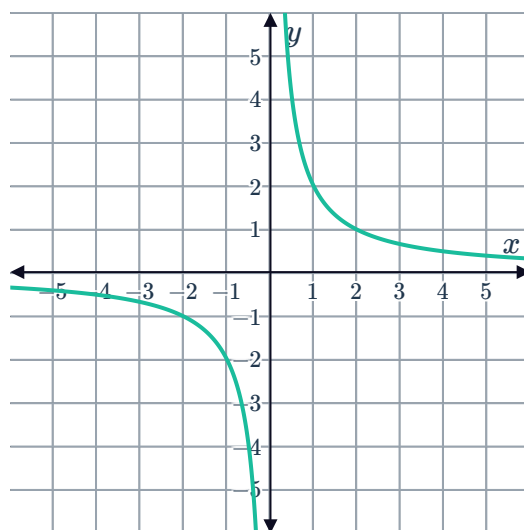
- 10 Use technology to graph the inverse relationship $y = \frac{6}{x}$.

- a How many x -intercepts does the graph have?
- b How many y -intercepts does the graph have?

Key features of a hyperbola

- 11 Consider the graph of $y = \frac{2}{x}$.

- a For $x > 0$, as x increases, what does y approach?
- b For $x > 0$, as x approaches 0, what does y approach?
- c State the equation of the vertical asymptote.
- d State the equation of the horizontal asymptote.
- e The graph has two axes of symmetry. State their equations.



12 The equation $y = -\frac{12}{x}$ represents an inverse relationship between x and y .



- a When $x = 3$, what is the value of y ?
- b If x is a positive value, is the corresponding y value positive or negative?
- c If x is a negative value, is the corresponding y value positive or negative?
- d In which quadrants does the graph of $y = \frac{-12}{x}$ lie?

13 Consider the function $y = \frac{2}{x}$.

- a Complete the table of values.

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y									

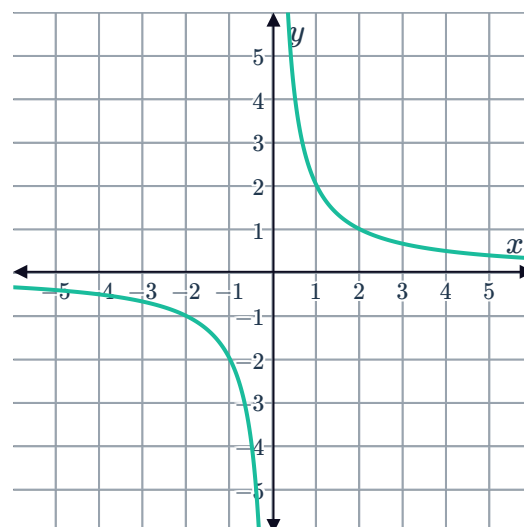
- b For what value of x is the function undefined?
- c Rewrite the equation to make x the subject.
- d For what value of y is the function undefined?

- e Below is the graph of $y = \frac{2}{x}$.

Complete the following statement:

As x approaches 0 from the right, the function value approaches ∞ . As x approaches 0 from the left, the function value approaches $-\infty$.

As x approaches ∞ and $-\infty$, y approaches 0. This is called the limiting value of the function.

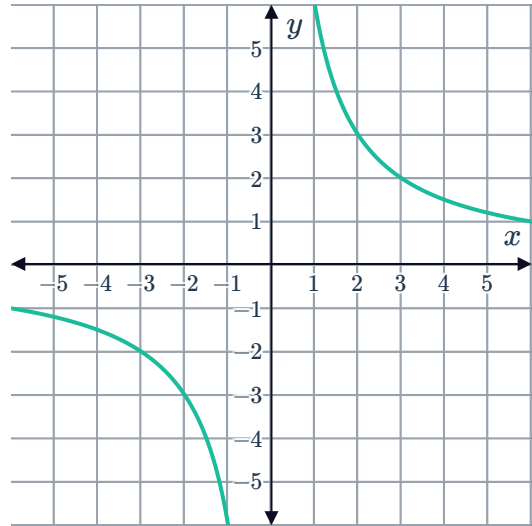


14 Consider the function $y = -\frac{5}{x}$.

- a For what value of x is the function undefined?
- b As x approaches 0 from the positive side, what does y approach?
- c As x approaches 0 from the negative side, what does y approach?
- d As x approaches ∞ , what does y approach?
- e As x approaches $-\infty$, what does y approach?



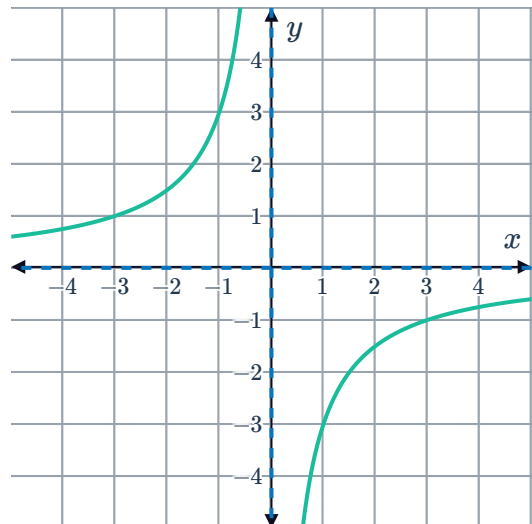
- 15 Consider the hyperbola shown:
Complete the statement:
"Every point (x, y) on the hyperbola is such
that $xy = 1$."



- 16 Determine whether the following relationships can be modelled by a function of the form $xy = a$?
- a The relationship between the number of people working on a job and how long it will take to complete the job.
 - b The relationship between the number of sales and the amount of revenue.
 - c The relationship between height and weight.

- 17 Consider the function graphed below:

- a What is the equation of the vertical asymptote?
- b What is the equation of the horizontal asymptote?
- c State the domain of the function.
- d State the range of the function.



- 18 For functions of the form $f(x) = \frac{k}{x}$, where k is a constant, the domain is always $(-\infty, 0) \cup (0, \infty)$. What does this mean for the function when $x = 0$?

19 Consider the function $y = \frac{1}{x}$.



- a For what value of x is the function undefined?
- b What is the domain of $y = \frac{1}{x}$?
- c For what value of y is the function underfined?
- d What is the range of $y = \frac{1}{x}$?

20 Patricia determined the range of the function $y = \frac{2}{x}$ is $(-\infty, 0) \cup (0, \infty)$.

By filling in the gaps below, complete her reasoning.

Looking at $y = \frac{2}{x}$, notice that the numerator $\mathbb{I}$ is non-zero, and so $\frac{2}{x}$ can never be equal to $\mathbb{I}$.
 Another approach is to rearrange the equation and make x the subject. Then we get $x = \mathbb{I}$.
 In this form we can see that the denominator $\mathbb{I}$ cannot be equal to $\mathbb{I}$.

21 Consider the functions $y = \frac{4}{x}$ and $y = \frac{2}{x}$.

- a When $x = 2$, what is the value of y if $y = \frac{4}{x}$?
- b When $x = 2$, what is the value of y if $y = \frac{2}{x}$?
- c Which graph lies further away from the axes?
- d For hyperbolas of the form $y = \frac{k}{x}$, as the k value increases, what happens to the graph?

Graphs of hyperbolas

22 Consider the inverse variation equation $y = \frac{6}{x}$.

- a Complete the following tables of values:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

- b Plot the points from the table of values on a cartesian plane.

23 Ursula wants to sketch the graph of $y = \frac{7}{x}$, but knows that it will look similar to many other hyperbolas.

What can she do to the graph to show that it is the hyperbola $y = \frac{7}{x}$, rather than any other hyperbola of the form $y = \frac{k}{x}$?

24 For each of the following functions:



i Complete a table of values of the form:

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

ii Sketch the graph of the function.

iii State which quadrants the function lies in.

a $y = \frac{2}{x}$

b $y = -\frac{1}{x}$

c $y = -\frac{1}{4x}$

Applications

25 The relationship between the current, C , (in amperes) and resistance, R , (in ohms) in an electrical circuit is given by:

$$C = \frac{240}{R}$$

where the voltage provided to the circuit is 240 V.

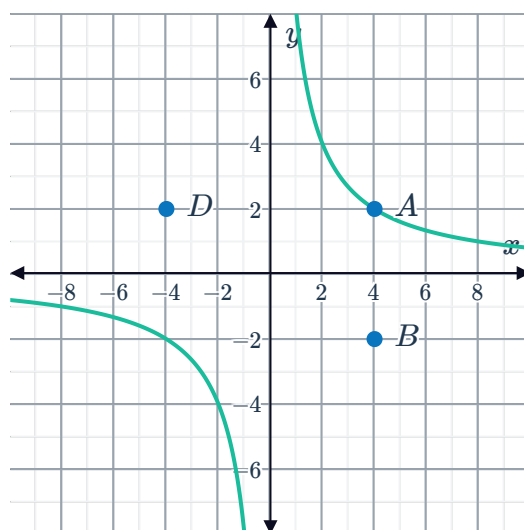
a Graph this function for $0 \leq R \leq 50$.

b What happens to the current as the resistance increases?

26 Consider the hyperbola that has been graphed. Points $A(4, 2)$, B , $C(-4, k)$ and D form the vertices of a rectangle.

a Find the value of k .

b Hence find the area of the rectangle with vertices $ABCD$.



- 27** Boyle's law describes the relationship between  pressure and volume of a gas of fixed mass under constant temperature. The pressure for a particular gas can be found using

$$P = \frac{6000}{V}$$

where P has units kg/cm^2 and V has units cm^3 .

- a** Graph the relationship $P = \frac{6000}{V}$ for $0 \leq V \leq 2000$.
 - b** What is the pressure if the volume is 1 cm^3 ?
 - c** What happens to the pressure as the volume increases?
- 28** The time it takes a commuter to travel 100 km depends on how fast they are going. We can write this using the equation $t = \frac{100}{S}$ where S is the speed in km/h and t is the time taken in hours.
- a** Graph the relationship $t = \frac{100}{S}$.
 - b** What is the time taken if the speed travelled is $10 \text{ km}/\text{h}$?
 - c** What is the time taken if the speed travelled is $50 \text{ km}/\text{h}$?
 - d** If we want the travel time to decrease, what must happen to the speed of travel?